Quantification: I534_z11c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	26
interior lacuna count	23
frame-edge lacuna count	3

Interior lacunae only: mean, median and sample SD (n = 23)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2954.39	2001	2699.77	px^2
major_axis_length_px	lacuna shape	84.9561	74.781	33.0167	px
minor_axis_length_px	lacuna shape	45.7378	37.2141	22.5561	px
aspect_ratio	lacuna shape	2.0152	1.9891	0.6188	unitless
eccentricity	lacuna shape	0.8185	0.8644	0.1167	unitless
solidity	lacuna shape	0.7412	0.724	0.0987	unitless
roots_count	headline	11.7826	11	5.8617	count
ring_length_r30_px	headline	321.913	308	156.145	px
ring_length_r60_px	reported	615.174	619	312.829	px
owned_length_px	ownership-dependent	820.833	633.985	659.125	px
edge_count	ownership-dependent	41.8696	31	31.4676	count
mean_edge_length_px	ownership-dependent	19.632	19.2965	4.0008	px
perimeter_px	normalised	281.136	270.635	124.881	px
ring_area_r30_px2	normalised	9267.43	8169	3530.85	px^2
ring_area_r60_px2	normalised	19156.3	17028	7217.71	px^2
in_frame_fraction_r30	normalised	0.9858	1	0.0341	unitless
in_frame_fraction_r60	normalised	0.9668	1	0.0743	unitless
ring_density_r30	normalised	0.0346645	0.0340687	0.0117903	px^-1
ring_density_r60	normalised	0.0317451	0.0337161	0.0108842	px^-1
roots_per_100px_perimeter	normalised	4.2166	4.1848	1.2215	per 100 px
ring_attached_length_r30_px	network v2	298.957	255	159.325	px
ring_attached_length_r60_px	network v2	505.043	426	323.182	px
ring_length_w_r30_px	network v2	365.224	349.22	177.454	px
ring_length_w_r60_px	network v2	697.989	683.931	355.975	px
sholl_crossings_r10	network v2	7.3478	7	4.1518	count
sholl_crossings_r20	network v2	7.913	7	4.295	count
sholl_crossings_r30	network v2	8.6522	8	4.5288	count
loop_count	loops, report only	2.1739	1	3.0399	count
loop_edge_share	loops, report only	0.1369	0.0667	0.1419	share
ring_width_mean_r30_px	width	7.6435	7.3149	1.3283	px

Per field
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measure	value	units
analysed_area_px2	775235	px^2
total_skeleton_length_px	23791	px
canalicular_length_density_per_px	0.0306888	px^-1
junction_count	552	count
junction_density_per_px2	0.00071204	px^-2
lacuna_count	26	count

lacunae_per_px2	3.354e-05	px^-2
skeleton_component_count	259	count
mean_component_length_px	91.8571	px
median_component_length_px	26	px
field_length_density_w_per_px	0.0348375	px^-1
field_density_without_flagged_per_px	0.0246377	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	104	px
unattached_component_count	7	count
canalicular_length_density_attached_per_px	0.0306888	px^-1
field_length_density_with_unattached_per_px	0.0308229	px^-1
excluded_area_px2	193353	px^2
analysed_field_area_px2	855223	px^2
loop_count	63	count
loop_edge_share	0.1567	share
edge_count	1468	count

Canalicular width, the new measure

measure	value	units
width_mean_px	8.1351	px
width_median_px	8.2462	px
width_p10_px	4	px
width_p90_px	11.6619	px
width_pixels_used	24353	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	5018	113.612	66.0072	1.7212	0.8139	0.7134	-1.5671	33.3908	243.731	True
2	1429	75.9455	34.6321	2.1929	0.89	0.5788	0.9821	47.7124	162.336	False
3	1748	63.7004	52.4126	1.2154	0.5683	0.625	1.3536	36.7677	399.808	False
4	2001	74.781	36.9486	2.0239	0.8694	0.794	-1.1924	101.66	153.378	False
5	4592	90.8676	67.3599	1.349	0.6712	0.8423	-1.2953	137.617	765.059	False
6	902	70.1443	23.2471	3.0173	0.9435	0.6677	-1.2472	150.552	535.413	False
7	2333	67.9441	45.1639	1.5044	0.7471	0.9554	-0.4874	234.38	1000.91	True
8	11887	168.982	95.145	1.7761	0.8264	0.8606	-1.2985	286.803	681.158	False
9	697	50.8222	20.9969	2.4205	0.9107	0.7383	1.293	287.073	212.017	False
10	3532	87.4182	55.1855	1.5841	0.7756	0.7638	1.235	329.842	344.026	False
11	4129	107.816	50.0459	2.1543	0.8857	0.9077	-1.34	378.661	498.241	False
12	889	60.4502	27.413	2.2052	0.8913	0.6247	-1.2388	363.547	49.8526	False
13	1735	68.9793	37.2098	1.8538	0.842	0.6893	-0.903	463.884	594.474	False
14	751	42.1527	25.0435	1.6832	0.8044	0.8632	-1.1817	459.581	482.401	False
15	5417	114.374	70.6956	1.6178	0.7861	0.6168	1.316	527.301	55.7192	True
16	3037	99.839	58.2931	1.7127	0.8118	0.571	1.4007	558.019	301.395	False
17	4073	115.2	49.8465	2.3111	0.9015	0.7391	0.3504	603.966	867.145	False
18	2337	72.0848	52.1015	1.3835	0.6911	0.6974	-1.4357	607.677	651.435	False
19	1679	74.0209	37.2141	1.9891	0.8644	0.724	-0.9638	634.934	701.784	False
20	7591	130.597	95.8386	1.3627	0.6793	0.6789	-0.2363	683.477	82.1523	False
21	4140	142.999	43.9592	3.253	0.9516	0.7236	-1.1238	670.561	384.423	False
22	554	30.7716	24.9383	1.2339	0.5858	0.895	1.0673	729.574	697.603	False

23	640	57.661	17.8443	3.2313	0.9509	0.7018	-0.3969	793.93	358.258	False
24	2036	78.7104	35.4441	2.2207	0.8929	0.8879	-1.002	828.682	94.3777	False
25	1332	74.3708	26.5147	2.8049	0.9343	0.75	-1.2358	829.074	388.601	False
26	6240	115.676	84.3355	1.3716	0.6844	0.7226	-1.0938	881.946	274.251	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	18	377	483	570.174	30	19.0058	6.3848
2	11	263	349	288.108	21	13.7194	6.2936
3	10	374	619	893.583	52	17.1843	7.2125
4	11	308	632	218.48	17	12.8518	6.588
5	17	451	750	1093.41	58	18.8518	7.3389
6	10	405	914	1060.58	60	17.6763	7.1077
7	6	176	413	620.132	28	22.1476	8.4729
8	26	700	1276	2421.78	120	20.1815	7.9982
9	7	227	478	589.902	31	19.0291	8.0294
10	17	550	1111	2410.82	114	21.1475	8.3002
11	12	356	676	848.352	51	16.6344	8.1886
12	8	210	361	309.635	16	19.3522	6.0559
13	13	396	709	1148.47	47	24.4356	7.3149
14	5	157	263	220.338	10	22.0338	8.1952
15	20	380	538	743.169	36	20.6436	5.7264
16	21	442	919	1427.94	74	19.2965	7.328
17	14	422	1101	1686.2	76	22.1868	7.0559
18	16	339	648	858.725	45	19.0828	7.2767
19	11	196	391	381.777	16	23.861	6.7916
20	21	552	923	945.406	47	20.115	7.1878
21	11	247	400	344.309	25	13.7723	7.0648
22	5	185	433	633.985	28	22.6423	8.1872
23	1	36	63	60.8701	2	30.435	13.0026
24	8	233	486	472.304	20	23.6152	8.1488
25	6	119	209	157.468	10	15.7468	7.2164
26	10	236	438	406.735	23	17.6841	7.9169

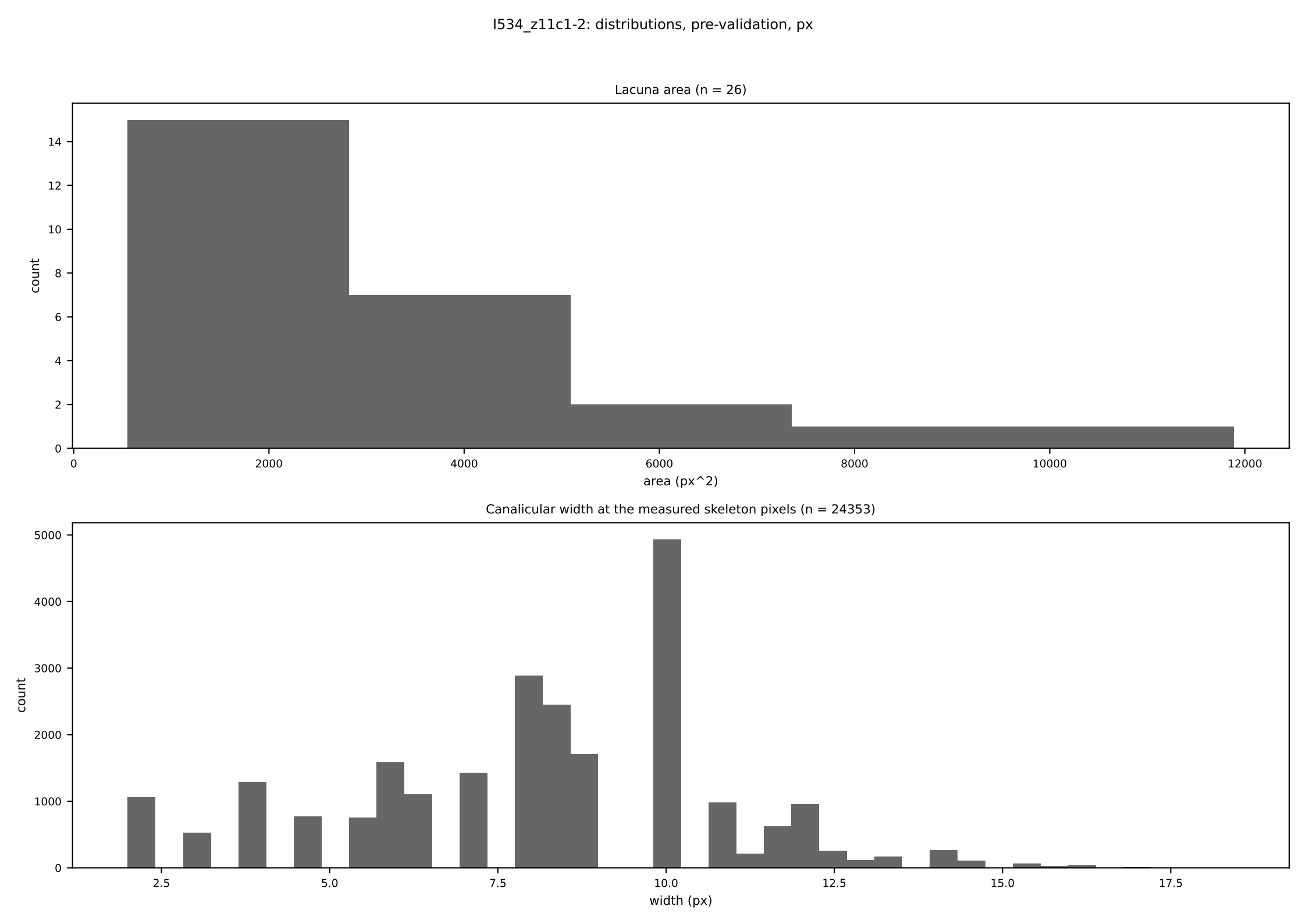
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	494.351	8231	12179	0.7421	0.6708	0.0458025	0.0396584	3.6411
2	282.291	4862	7142	0.9455	0.8134	0.054093	0.0488659	3.8967
3	270.635	8169	16049	0.8695	0.7737	0.0457828	0.0385694	3.695
4	236.936	6970	15047	1	1	0.0441894	0.0420017	4.6426
5	341.505	11247	24451	1	1	0.0400996	0.0306736	4.978
6	176.167	7690	20930	1	1	0.0526658	0.0436694	5.6764
7	199.782	5730	14247	0.6804	0.6335	0.0307155	0.0289886	3.0033
8	548.257	17191	33534	1	1	0.040719	0.0380509	4.7423
9	136.125	6663	16736	1	1	0.0340687	0.0285612	5.1423
10	297.706	11030	23297	1	1	0.049864	0.0476885	5.7103
11	305.605	10599	17028	1	1	0.0335881	0.0396993	3.9266
12	191.167	7524	16666	0.9189	0.7962	0.0279107	0.0216609	4.1848
13	238.451	9266	19320	1	1	0.0427369	0.0366977	5.4519
14	122.953	5564	11136	1	1	0.0282171	0.0236171	4.0666
15	490.399	11727	20563	0.8061	0.7376	0.0324038	0.0261635	4.0783
16	378.475	12733	27257	1	1	0.0347129	0.0337161	5.5486
17	373.546	12922	30830	1	1	0.0326575	0.035712	3.7479
18	290.877	7526	16187	1	1	0.0450439	0.0400321	5.5006
19	233.38	6410	12295	1	1	0.0305772	0.0318015	4.7133

20	555.955	15261	25089	0.9401	0.8614	0.0361706	0.036789	3.7773
21	383.889	12892	25468	1	1	0.0191592	0.015706	2.8654
22	97.9828	5477	13592	1	1	0.0337776	0.031857	5.1029
23	139.054	4710	7157	1	1	0.00764331	0.00880257	0.7191
24	222.693	8786	20073	1	0.9917	0.0265195	0.0242116	3.5924
25	196.066	6012	12715	1	1	0.0197937	0.0164373	3.0602
26	446.416	13647	28597	1	1	0.0172932	0.0153163	2.2401

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	372	459	430.09	545.931	10	9	4
2	255	290	293.865	392.291	9	5	3
3	370	567	416.078	683.931	7	9	6
4	206	206	349.22	712.571	7	6	8
5	431	661	514.86	850.524	11	11	13
6	393	722	460.505	1046.55	9	11	10
7	172	413	186.498	453.321	4	4	5
8	680	1228	778.357	1436.54	17	17	17
9	195	334	264.522	549.245	5	6	6
10	536	1081	626.073	1266.22	12	16	19
11	342	628	411.505	782.21	9	10	14
12	147	171	244.421	419.931	5	3	5
13	376	681	457.889	827.737	6	8	9
14	134	162	178.539	295.037	4	4	5
15	344	369	437.132	627.512	8	4	4
16	433	779	497.961	1034.99	13	12	11
17	420	930	480.889	1234.69	8	11	13
18	339	633	393.605	742.198	6	8	7
19	185	222	214.823	439.233	3	5	4
20	503	802	639.441	1067.12	15	14	13
21	223	306	268.924	433.25	4	4	8
22	166	315	205.125	496.274	4	6	6
23	10	10	38.3848	69.1127	0	1	2
24	229	426	268.794	554.144	6	6	8
25	119	159	135.397	231.823	2	2	3
26	184	303	260.966	488.132	7	7	9


I534_z11c1-2: distributions, pre-validation, px
Lacuna area (n = 26)
count
area (px^2)
0
2
4
6
8
10
12
14
0
2000
4000
6000
8000
10000
12000
Canalicular width at the measured skeleton pixels (n = 24353)
count
width (px)
0
1000
2000
3000
4000
5000
2.5
5.0
7.5
10.0
12.5
15.0
17.5